

Nik Abdul Aqmal Bin Zunaidi

+6017-9716757 | nikabdulaqmal@gmail.com | github.com/nickaqmal02

PROFESSIONAL SUMMARY

Final-year Computer Science student at UPNM (CGPA 3.79) building end-to-end ML systems from data pipelines (BERTopic, XLM-RoBERTa) to production interfaces (Django, DRF). Built a personal RAG agent to explore retrieval-augmented generation pursuing internship to apply this in a real engineering team where I can contribute to AI/ML product development.

EDUCATION

- **STPM | SMK ISMAIL PETRA | [SCIENCE STREAM] | 3.92**
- **DEGREE | UPNM | [BC. COMPUTER SCIENCE W. HONOURS] | 3.81**

PROJECTS

Sentiri - Autism Therapy Chat Analytics Platform | Python, Django, Django REST Framework, BERTopic,Sep 2025 – Present
XLM-RoBERTa, PostgreSQL

- Built hybrid ML pipeline combining supervised keyword mapping with unsupervised BERTopic clustering to automate topic assignment across 400+ WhatsApp messages, enabling end-to-end categorization from data ingestion through classification.
- Built a RAG-powered treatment Q&A chatbot implementing ChromaDB for vector store with multilingual sentence embeddings, cross-encoder reranking and celery based document ingestion upload.
- Built a LangChain reservation chatbot with redis persisted conversation memory, integrated with DRF APIs to complete bookings directly from chat.
- Built end-to-end multilingual sentiment classification system by fine-tuning XLM-RoBERTa with integrated BERTopic modeling and custom seed-based supervised mapping, delivering accurate positive/negative/neutral categorization across diverse language datasets.
- Github repository: **github.com/nickaqmal02/finalyearenginefulleffort**

ApexAgent - Native macOS AI Assistant | Swift, SwiftUI, Ollama, GLM API | Oct 2026 - Present

- Built a Swift-native agent with a command router and online/offline fallback, routing between local Ollama (qwen3:8b) and a cloud LLM API for uninterrupted without network dependency.
- With voice pipeline using (Deepgram STT + AVAudioEngine) in active development.
- Github repository: **github.com/nickaqmal02/ApexAgent**

PersonalRAG - Retrieval Augmented Generation System | Python, LangChain, ChromaDB, Groq, SentenceTransformers, Textual

- Built end to end RAG pipeline (ingestion -> chunking -> embedding -> retrieval -> reranking -> generation) with ChromaDB vector store, Sentence Transformers embeddings, and Groq LLM.
- Implemented modular package architecture (core/pipeline/llm/config/tui separation).
- Added reranking stage to improve retrieval quality over baseline top-K.
- Github repository: **github.com/nickaqmal02/PersonalRAG**

Django E-Commerce Platform with Stripe Integration | Django, PostgreSQL, StripeAPI, Tailwind CSSSep 2026 – Oct 2026

- Shipped a full-stack e-commerce application enabling end-to-end transactions through Stripe Checkout integration, webhook handling, and automated PDF receipt generation with WeasyPrint, deployed to production on Render with gunicorn and WhiteNoise static serving for seamless customer checkout experience.
- Github repository: **github.com/nickaqmal02/django_ecommerce_stripe**

RESEARCH

NLP Sentiment Analysis Study | UPNM | Under Review 2026 | co-authored comparative study showing fine-tuned BERT outperforms lexicon-based sentiment methods.

CERTIFICATIONS

MATLAB Certified

SKILLS

Programming Languages: Python, PHP, Javascript, MATLAB

Frameworks & Libraries: Django, Laravel, Django REST Framework, FastAPI, ReactJS, Textual, Poetry, LangChain, Scikit-learn

AI & Data Science: NLP, BERT, XLM-RoBERTa, BERTopic, Computer Vision, YOLOv8, OpenCV, Sentiment Analysis, Retrieval Augmented Generation, Prompt Engineering

Databases: PostgreSQL, MySQL, SQLite, ChromaDB

Developer Tools & Environment: Neovim, Ubuntu/Linux, Bash/CLI, Git, Github, Docker

APIs & Services: Claude API, DeepSeek API, Groq LLM, StripeAPI, WeasyPrint

INTERESTS

· Agentic AI & LLM research · DevOps · Linux · Neovim · Terminal Native · English classic literature · AI Automation · Engineering Performance · Applied Mathematics

REFERENCES

Prof. Associate Dr. Muslihah Binti Wook
Lecturer, Defence Science Technology Faculty, UPNM
muslihah@upnm.edu.my | +6012-5628252